

Aim: Linux Commands: Exploring the Raspbian.

1.ls -l:

```

pi@pi: ~
File Edit Tabs Help

pi@pi:~$ ls -l
total 404
-rw-r--r-- 1 pi pi 128917 Feb 2 10:19 2026-02-02-101915_1366x768_scrot.png
drwxr-xr-x 3 pi pi 4096 Oct 3 11:15 4digittime
drwxr-xr-x 3 pi pi 4096 Oct 3 11:15 4digitTime
drwxr-xr-x 3 pi pi 4096 Oct 4 13:16 7timesegment
-rw-r--r-- 1 pi pi 444 Sep 25 12:49 aarti.py
drwxr-xr-x 9 pi pi 4096 Sep 4 2024 Adafruit_Python_ADS1x15
drwxr-xr-x 10 pi pi 4096 Sep 6 2023 Adafruit_Python_ADS1x15-temp
drwxr-xr-x 9 pi pi 4096 Sep 27 2023 Adafruit_Python_CharLCD
-rw-r--r-- 1 pi pi 283 Mar 8 2025 blinkblink.py
-rw-r--r-- 1 pi pi 504 Jul 19 2023 blinking_led.py
-rw-r--r-- 1 pi pi 279 Oct 20 2023 blinkled.py
-rw-r--r-- 1 pi pi 997 Aug 20 13:26 blink.pyc
drwxr-xr-x 2 pi pi 4096 Aug 20 2020 Bookshelf
drwxr-xr-x 3 root root 4096 Jan 25 2023 build
-rw-r--r-- 1 pi pi 320 Oct 4 2023 cameradoc.py
-rw-r--r-- 1 pi pi 199 May 8 2023 Camera.py
-rw-r--r-- 1 pi pi 236 Oct 4 2023 cam.py
-rwxr-xr-x 1 root root 1901 Oct 5 2023 cmdline.txt
drwxr-xr-x 4 pi pi 4096 Aug 25 12:22 CN_Grp-1
drwxr-xr-x 3 pi pi 4096 Feb 15 2024 dem
drwxr-xr-x 7 pi pi 4096 Oct 3 14:56 Desktop

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2.ls -h, ls -t, ls -r, pwd:

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pi@pi: ~
File Edit Tabs Help

pi@pi:~$ ls -h
.bash_history      cmdline.txt      .idlerc         max7219         .pp_backup      SYCS            .thumbnails
.bash_logout      .CN_Grp-1      image.jpg~     Music          practical3      SYCS01.txt     time.py
.bashrc           .config        image.py       new1.py        'practical 8.py' SYCS19         tRY.py
pi@pi:~$ ls -t
2026-02-02-102013_1366x768_scrot.png  blinking_led.py  dem              krishna          newtel.py       Rane.py        sycs2          time.py
2026-02-02-102405_1366x768_scrot.png  blink.pyc       Desktop         lcd              op.html         script.js      sycs65.py     tRY.py
4digittime                             blink.pyc       dist            LED-Grid_Yash.py orio.py         send.py       TBomb          UNKNOWN.egg-info
4digitTime                             Bookshelf       Documents      led.py          led.py          oscilloscopedemo.py seven         tele4led.py   Videos
7timesegment                           build           Downloads      led.py          LED.py          p11.py        Pictures       Telegram
aarti.py                               cameradoc.py    fourledblink.py LED.py          mamta           image.py      Pictures       telegram4led.py
Adafruit_Python_ADS1x15                 Camera.py       image.jpg~     manta           max7219         'practical 8.py' sycs          telegram.py
Adafruit_Python_ADS1x15-temp             cam.py         image.py       'IOT Pracs'     kit3.py         'Public'      SYCS          tel.py
Adafruit_Python_CharLCD                 cmdline.txt     'IOT Pracs'   kit3.py         new1.py         py.py         SYCS01.txt    Templates
blinkblink.py                           CN_Grp-1       kit3.py       kit3.py         new1.py         py.py         SYCS19        test.jpg~
pi@pi:~$ ls -t
2026-02-02-102433_1366x768_scrot.png  'IOT Pracs'     shyam.py        SYCS01.txt      blinkled.py     fourledblink.py  Soham.py
2026-02-02-102405_1366x768_scrot.png  4digitTime     led.py         'practical 8.py' cameradoc.py    blinking_led.py  newtel.py
2026-02-02-102013_1366x768_scrot.png  4digitTime     SVCS19        Telegram        oscilloscopedemo.py cam.py          LED.py
Downloads                               Rane.py        SYCS          kit3.py         Adafruit_Python_ADS1x15 Adafruit_Python_CharLCD Led.py
tRY.py                                 orio.py        sycs2         oscilloscopedemo.py Adafruit_Python_ADS1x15-temp Adafruit_Python_ADS1x15-temp Krishna
7timesegment                           aarti.py      blinkblink.py  Adafruit_Python_ADS1x15 tel.py          tel.py          Public
time.py                                p11.py        LED-Grid_Yash.py Adafruit_Python_ADS1x15 tel.py          telegram.py      Templates
Desktop                                CN_Grp-1      op.html       Documents       new1.py         tele4led.py     dist          Videos
py.py                                  send.py       lcd           dem             telegram.py      tele4led.py     build        Bookshelf
seven                                  blink.pyc     test.jpg~     script.js       telegram.py      UNKNOWN.egg-info max7219
practical3                             sycs65.py     sycs          telegram.py
pi@pi:~$ ls -r
Videos      TBomb      script.js  newtel.py  kit3.py    cmdline.txt  Adafruit_Python_ADS1x15-temp
UNKNOWN.egg-info sycs65.py  Rane.py   new1.py    'IOT Pracs' cam.py       Adafruit_Python_ADS1x15
tRY.py       sycs2     py.py     Music      image.py   Camera.py   aarti.py
time.py      SVCS19   'practical 8.py' max7219    image.jpg~ cameradoc.py 7timesegment
test.jpg~    SVCS01.txt 'practical 8.py' LED.py     fourledblink.py build
Templates    sycs     Pictures  LED.py     Downloads  Bookshelf
tel.py       sycs2    p11.py   led.py     Documents  blink.pyc
telegram.py  Soham.py oscilloscopedemo.py led.py     dist       blinkled.py
telegram4led.py shyam.py  led.py   LED-Grid_Yash.py Desktop    blinkblink.py
Telegram      seven     orio.py  lcd        dem         blinkblink.py
tele4led.py  send.py  op.html  krishna    CN_Grp-1   Adafruit_Python_CharLCD
pi@pi:~$ pwd
/home/pi
pi@pi:~$

```

3.mkdir directory_name, ls, cd directory_name, cd, mkdir -p directory1/directory2, rmdir directory_name, rmdir -v directory_name:

```

pi@pi:~$ mkdir jay
pi@pi:~$ ls
2026-02-02-102013_1366x768_scrot.png  Adafruit_Python_CharLCD  dem          lcd          oscilloscopedemo.py  Soham.py  tel.py
2026-02-02-102405_1366x768_scrot.png  blinkblink.py            Desktop     LED-Grid_Yash.py p11.py       sycs       Templates
2026-02-02-102433_1366x768_scrot.png  blinkled.py             dist        led.py       Pictures      SVCS       test.jpg~
2026-02-02-102459_1366x768_scrot.png  blink.pyc               Documents   LED.py       'practical3  SYCS01.txt time.py
2026-02-02-102516_1366x768_scrot.png  blink.pyc               Downloads   LED.py       'practical 8.py' SYCS19    tRY.py
2026-02-02-102544_1366x768_scrot.png  Bookshelf               fourledblink.py mamta        Public       sycs2     UNKNOWN.egg-info
4digittime                             build                   image.jpg~  max7219     py.py       sycs65.py Videos
4digitTime                             cameradoc.py            image.py    new1.py     py.py       TBomb
7timesegment                           Camera.py               'IOT Pracs' newtel.py   script.js   tele4led.py
aarti.py                               cam.py                  kit3.py     op.html     seven       telegram4led.py
Adafruit_Python_ADS1x15                 cmdline.txt             CN_Grp-1   krishna     orio.py     telegram.py
Adafruit_Python_ADS1x15-temp             CN_Grp-1
pi@pi:~$ cd jay
pi@pi:~/jay$ cd
pi@pi:~$ cd jay
pi@pi:~/jay$ mkdir -p rohit/zameer
pi@pi:~/jay$ ls
rohit
pi@pi:~/jay$ rmdir rohit
rmdir: failed to remove 'rohit': Directory not empty
pi@pi:~/jay$ cd rohit
pi@pi:~/jay/rohit$ ls
zameer
pi@pi:~/jay/rohit$ rmdir zameer
pi@pi:~/jay/rohit$ ls
pi@pi:~/jay/rohit$ cd
pi@pi:~$ cd jay
pi@pi:~/jay$ rmdir -v rohit
rmdir: removing directory, 'rohit'
pi@pi:~/jay$ ls
pi@pi:~/jay$

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pi@pi:~/jay $ rmdir -p jay
rmdir: failed to remove 'jay': No such file or directory
pi@pi:~/jay $ cd
pi@pi:~$ rmdir -p jay
pi@pi:~$ ls
2026-02-02-102013_1366x768_scr0t.png  4digittime      cameradoc.py      image.py           newl.py           script.js          tele4led.py
2026-02-02-102405_1366x768_scr0t.png  7timesegment    Camera.py          'IOT Pracs'       newtel.py         send.py            Telegram
2026-02-02-102433_1366x768_scr0t.png  aarti.py        cam.py            kit3.py            op.html           seven              telegram4led.py
2026-02-02-102459_1366x768_scr0t.png  Adafruit_Python_ADS1x15  cmdline.txt      lcd                orio.py           shyam.py          telegram.py
2026-02-02-102516_1366x768_scr0t.png  Adafruit_Python_ADS1x15-temp  CN_Grp-1        led                oscilloscopedemo.py  p11.py           tel.py
2026-02-02-102544_1366x768_scr0t.png  Adafruit_Python_CharLCD    dem             LED_Grid_Yash.py  p11.py           Pictures          Templates
2026-02-02-102628_1366x768_scr0t.png  blinkblink.py    Desktop          led.py            Pictures          SVCs              test.jpg~
2026-02-02-102743_1366x768_scr0t.png  blinking_led.py  dist            Led.py            SVCs01.txt       SVCs              time.py
2026-02-02-102827_1366x768_scr0t.png  blinkled.py     Documents        LED.py            SVCs19           tRY.py            trY.py
2026-02-02-102847_1366x768_scr0t.png  blink.pyc       Downloads        manta             Public            sycs2             UNKNOWN.egg-info
2026-02-02-102913_1366x768_scr0t.png  Bookshelf       fourledblink.py  max7219           py.py            sycs65.py         Videos
4digittime                          build            image.jpg~       Music              Rane.py          TBomb
pi@pi:~$ cat f.txt
cat: f.txt: No such file or directory
pi@pi:~$ touch f.txt
pi@pi:~$ cat f.txt
pi@pi:~$ nano f.txt
pi@pi:~$ cat > f.txt
hello world
Hi I am Jay, Rohit, Zameer
pi@pi:~$ cat f.txt
hello world
Hi I am Jay, Rohit, Zameer
pi@pi:~$ head f.txt
hello world
Hi I am Jay, Rohit, Zameer
pi@pi:~$ tail f.txt
hello world
Hi I am Jay, Rohit, Zameer
pi@pi:~$ tac f.txt
Hi I am Jay, Rohit, Zameer
hello world
pi@pi:~$

```

```

pi@pi:~$ df
Filesystem      1K-blocks    Used Avail Use% Mounted on
/dev/root        14990416  5871932   8458736  41% /
devtmpfs         413012      0    413012   0% /dev
tmpfs            446292    30748   415544   7% /dev/shm
tmpfs            446292    17260   429032   4% /run
tmpfs            5120        4    5116    1% /run/lock
tmpfs            446292      0    446292   0% /sys/fs/cgroup
/dev/mmcblk0p1   258095    49403   208693   20% /boot
tmpfs            89256      8    89248    1% /run/user/1000

pi@pi:~$ free
              total        used        free      shared  buff/cache   available
Mem:           892584      486052      25292      38644      32940      315848
Swap:          102396       98272        4124

pi@pi:~$ hostname
pi

pi@pi:~$ touch t.txt

pi@pi:~$ ls
2026-02-02-102013 1366x768_scr0t.png  2026-02-02-103621 1366x768_scr0t.png  Camera.py          lcd                  'practical 8.py'   tele4led.py
2026-02-02-102405 1366x768_scr0t.png  2026-02-02-103627 1366x768_scr0t.png  cam.py            LED_Grid_Yash.py   Public            Telegram
2026-02-02-102433 1366x768_scr0t.png  4digitTime       cmdline.txt         led.py            led.py              py.py            telegram4led.py
2026-02-02-102459 1366x768_scr0t.png  4digitTime       CN_Grp-1            Led.py            Rane.py             telegram.py       tel.py
2026-02-02-102516 1366x768_scr0t.png  7timesegment     dem                 LED.py            script.js           send.py          Templates
2026-02-02-102544 1366x768_scr0t.png  aarti.py         Desktop             mamta             max7219             seven            test.jpg~
2026-02-02-102628 1366x768_scr0t.png  Adafruit_Python_ADSiX15  dist               Documents          Music               shyam.py         time.py
2026-02-02-102743 1366x768_scr0t.png  Adafruit_Python_ADSiX15-temp  Downloads          new1.py            newtel.py         Soham.py          tRV.py
2026-02-02-102847 1366x768_scr0t.png  Adafruit_Python_CharLCD  blinkblink.py     fourledblink.py   f.txt              op.html           orio.py           UNKNOWNS.egg-info
2026-02-02-102913 1366x768_scr0t.png  blinking_led.py   blinkled.py        image.jpg~         oscilloscopedemo.py  p11.py           sycs2             Videos
2026-02-02-103006 1366x768_scr0t.png  blink.pyc        Bookshelf          build              kit3.py            Pictures          sycs65.py        TBomb
2026-02-02-103425 1366x768_scr0t.png  cameradoc.py     krishna            practical3
pi@pi:~$

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pi@pi1:~$ mv f.txt q.txt
pi@pi1:~$ ls
2026-02-02-102013 1366x768 _scrot.png    2026-02-02-103621 1366x768 _scrot.png    cameradoc.py    led    'practical 8.py'    TBomb
2026-02-02-102405 1366x768 _scrot.png    2026-02-02-103627 1366x768 _scrot.png    Camera.py    LED-Grid_Yash.py    Public    teledled.py
2026-02-02-102413 1366x768 _scrot.png    2026-02-02-103652 1366x768 _scrot.png    cam.py    led.py    py.py    Telegram
2026-02-02-102459 1366x768 _scrot.png    4digittime    cml1ine.txt    led.py    q.txt    telegram4led.py
2026-02-02-102516 1366x768 _scrot.png    4digittime    CU Grp-1    LED.py    Rane.py    telegram.py
2026-02-02-102544 1366x768 _scrot.png    7Timesegment    manta    script.js    tel.py
2026-02-02-102628 1366x768 _scrot.png    aarti.py    Desktop    max7219    Templates
2026-02-02-102743 1366x768 _scrot.png    Adafruit_Python_ADS1x15    dist    Music    test.jpg-
2026-02-02-102827 1366x768 _scrot.png    Adafruit_Python_ADS1x15-temp    Documents    new1.py    shyam.py    time.py
2026-02-02-102847 1366x768 _scrot.png    Adafruit_Python_CharLCD    Downloads    newtel.py    Soham.py    tRY.py
2026-02-02-102913 1366x768 _scrot.png    blinkblink.py    fourledblink.py    op.html    sycs    t.txt
2026-02-02-103066 1366x768 _scrot.png    blinking_led.py    image.jpg-    orio.py    SYCS    UNKNOWN.egg-info
2026-02-02-103350 1366x768 _scrot.png    blinkled.py    image.py    oscilloscopedemo.py    SYCS01.txt    Videos
2026-02-02-103425 1366x768 _scrot.png    blink.pyc    p11.py    SYCS19    SYCS19
2026-02-02-103510 1366x768 _scrot.png    Bookshelf    kit3.py    Pictures    sycs2
2026-02-02-103615 1366x768 _scrot.png    build    krishna    practical3    sycs85.py
pi@pi1:~$

```

7.wc filename.txt, rm filename.txt, ping:

```

pi@pi:~$ wc q.txt
 2  8 39 q.txt
pi@pi:~$ rm t.txt
pi@pi:~$ ls
2026-02-02-102013_1366x768_scr0t.png  2026-02-02-103510_1366x768_scr0t.png  blinkblink.py  Documents  max7219  Rane.py  Telegram
2026-02-02-102405_1366x768_scr0t.png  2026-02-02-103615_1366x768_scr0t.png  blinking_led.py  Downloads  Music  script.js  telegram4led.py
2026-02-02-102433_1366x768_scr0t.png  2026-02-02-103621_1366x768_scr0t.png  blinkled.py  fourledblink.py  new1.py  send.py  telegram.py
2026-02-02-102459_1366x768_scr0t.png  2026-02-02-103627_1366x768_scr0t.png  blink.pyc  image.jpg~  newtel.py  seven  tel.py
2026-02-02-102516_1366x768_scr0t.png  2026-02-02-103652_1366x768_scr0t.png  Bookshelf  image.py  op.html  shyam.py  Templates
2026-02-02-102544_1366x768_scr0t.png  2026-02-02-103902_1366x768_scr0t.png  build  'IOT Pracs'  orio.py  Soham.py  test.jpg~
2026-02-02-102628_1366x768_scr0t.png  2026-02-02-104021_1366x768_scr0t.png  cameradoc.py  kit3.py  oscilloscopedemo.py  syscs  time.py
2026-02-02-102743_1366x768_scr0t.png  4digitTime  Camera.py  krishna  p11.py  SYCS  tRY.py
2026-02-02-102827_1366x768_scr0t.png  4digitTime  cam.py  lcd  Pictures  SYCS01.txt  UNKNOWN.egg-info
2026-02-02-102847_1366x768_scr0t.png  7timesegment  cmdline.txt  LED_Grid_Yash.py  'practical3  SYCS19  Videos
2026-02-02-102913_1366x768_scr0t.png  aarti.py  CN_Grp-1  led.py  'practical 8.py'  syscs2
2026-02-02-103006_1366x768_scr0t.png  Adafruit_Python_ADS1x15  dem  led.py  Public  syscs65.py
2026-02-02-103350_1366x768_scr0t.png  Adafruit_Python_ADS1x15-temp  Desktop  LED.py  py.py  TBomb
2026-02-02-103425_1366x768_scr0t.png  Adafruit_Python_CharLCD  dist  mamta  q.txt  tele4led.py
pi@pi:~$ ping
Usage: ping [-aAbBdDfHhLnOqrRUVvV64] [-c count] [-i interval] [-I interface]
           [-m mark] [-M pmtudisc_option] [-l preload] [-p pattern] [-Q tos]
           [-s packetsize] [-S sndbuf] [-t ttl] [-T timestamp_option]
           [-w deadline] [-W timeout] [hop1 ...] destination
Usage: ping -6 [-aAbBdDfHhLnOqrRUVvV] [-c count] [-i interval] [-I interface]
           [-l preload] [-m mark] [-M pmtudisc_option]
           [-N nodeinfo_option] [-p pattern] [-Q tclass] [-s packetsize]
           [-S sndbuf] [-t ttl] [-T timestamp_option] [-w deadline]
           [-W timeout] destination
pi@pi:~$

```

8.date, ifconfig, ps:

```

pi@pi:~$ date
Mon 02 Feb 2026 10:42:02 AM IST
pi@pi:~$ id
uid=1000(pi) gid=1000(pi) groups=1000(pi),4(adm),20(dialout),24(cdrom),27(sudo),29(audio),44(video),46(plugdev),60(games),100(users),105(input),109(netdev),997(gpio),998(i2c),999(spi)
pi@pi:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.3.16 netmask 255.255.255.0 broadcast 192.168.3.255
    inet6 fe80::12ed:8a22:2bad:9caa prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:bc:fa:6b txqueuelen 1000 (Ethernet)
    RX packets 127198 bytes 107615152 (102.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12512 bytes 2287846 (2.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 53 bytes 9574 (9.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 53 bytes 9574 (9.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether b8:27:eb:e9:af:3e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

pi@pi:~$ ps
  PID TTY          TIME CMD
 1962 pts/0    00:00:00 bash
 2219 pts/0    00:00:00 ps
pi@pi:~$

```